

# **SOR Assignment 4**

Sensors and Perception

Short Report

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**Date:** June 25, 2026

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## **1 How did you estimate distance?**

I used the depth camera available in the simulation. After YOLO detected the target object, I calculated the centre of its bounding box and read the corresponding value from the depth image. This depth value directly gave the estimated distance between the robot and the target object.

## **2 How does the search behaviour work?**

When the target object is not detected in the camera frame, the robot enters search mode. In this mode it continuously rotates in place, increasing the camera field of view until the target becomes visible. Once the target is detected, the search behaviour stops automatically.

## **3 How does the robot decide when to move and stop?**

After detecting the target, the robot first aligns itself so that the target is near the centre of the camera image. Once aligned, it moves forward while continuously updating the estimated distance. When the measured distance becomes smaller than the chosen stopping threshold, the robot publishes zero velocity commands and ends the mission.

## 4 Source Code GitHub Repository

All code modifications, simulation launch configurations, and scripts implemented for this assignment are hosted on GitHub. The complete working directory, commit history, and setup instructions can be accessed via the link below:

**GitHub Repository Link:** [Click here to view the Repository](#)